Unit 1 – Using Objects and Methods Name: **KEY**
Checkpoint Practice 1

B 1) Consider the following code segment.

```
System.out.print("PI ");
System.out.println();
System.out.println("AN ");
System.out.print("O ");
```

What is printed as a result of executing the code segment?

(A) PIANO

(B) PI
AN
O

(C) PI
ANO

(D) PIAN
O

B 2) Which of the following expressions is equal to 1?

(A) 10 % 5
(B) 11 % 5
(C) 10 / 5
(D) 11 / 5

A 3) What are the values of `a` and `b` after the following code executes?

```
int a = 10, b = 20;
a += b;
b += a;
b += a % 4;
a -= b / 11;
```

a	b
~~10~~	~~20~~
~~30~~	~~50~~
26	52

(A) a = 26 and b = 52
(B) a = 0 and b = 2
(C) a = 17 and b = 40
(D) a = 28 and b = 57

4) Given the following code segments, trace the value of the variable to determine what is printed.

a)

```
int x = 8;
x /= 4;
x *= 3;
System.out.println(x);
```

6

b)

```
int num = 5;
num++;
num /= 3;
double p = 2.5;
p *= num;
System.out.println(p);
```

5.0

5) Trace each code segment and write the value that will be printed.

a)

```
int a = 18;
int b = 5;
a %= b;
a += b * 2;
System.out.println(a);
```

13

b)

```
int p = 4;
int q = 6;
double t = 1.5;
p *= q - 2;
q += p % 3;
t = p + q * 2;
System.out.println(p + q + t);
```

53.0

c)

```
int a = 3;
a *= 4;
a -= 5;
a += 2;
System.out.println(a);
```

9

d)

```
int x = 9;
int y = 4;
x += y;
y = x - y;
x *= y;
System.out.println(x + y);
```

126

6) On the line next to the variable description, write the data type that would represent it the best.

String a) A variable used to store your dog's name

double b) A variable used to store the square root of 2

int c) A variable used to store your age

boolean d) A variable used to store the answer to a yes or no question

boolean
double
String
int

7) For the following expressions, add a single set of parentheses so you obtain the intended value.

```
int a = 1, b = 2, c = 3, d = 4;
```

a) Value: 2

```
a + b * c + d / b % 3

1 + 2 * (3 + 4) / 2 % 3
1 + 2 * 7 / 2 % 3
1 + 14 / 2 % 3
1 + 7 % 3
1 + 1
2
```

b) Value: 1

```
a - b * d % 4 + c / 3

1 - 2 * 4 % (4 + 3) / 3
1 - 2 * 4 % 7 / 3
1 - 8 % 7 / 3
1 - 1 / 3
1 - 0
1
```